

Muhammad Hamza

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Education

Indian Institute of Technology Kharagpur

Dual Degree (B.Tech + M.Tech), Specialization in Artificial Intelligence & Applications

Kharagpur, India

Aug 2023 – May 2028

CGPA: 8.62/10.00

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Numerical Methods, Database Management Systems

Research Interests

Scientific Machine Learning, Neural Operators, Multimodal Learning, Computer Vision, Efficient Deep Learning, Classical Machine Learning

Research Experience

University of Manchester × NASA 🌐

Manchester, UK

Research Intern @ Scientific ML (Supervisor: Prof. Anirbit Mukherjee)

Nov 2025 – Present

- Designed and implemented **DeepONet operators** on spherical domains, modeling heat PDE dynamics with **maximum harmonic mode $L_{\max} = 511$** , achieving stable spectral evolution and accurate eigenvalue decay.
- Engineered geometry-aware architectures combining **Graph CNN (HEALPix, ~196K nodes)** & **SIREN trunk networks**, mitigating spectral bias and enabling high-frequency function reconstruction on S^2 manifolds.
- Developed spectral-domain operator learning pipelines using **spherical harmonic transforms (SHT)**, ensuring alias-free representations up to **$L_{\max} = 511$** and preserving orthogonality for robust PDE generalization.
- Optimized scalable training pipelines via **stochastic spatial sampling (5K/196K nodes per step)**, significantly reducing the computational cost by $\sim 40\times$ while preserving convergence and operator fidelity.

Independent Research - Geometry-Aware Deep Learning Optimization 🌐

Independent Researcher

Feb 2026 – Present

- Developed **Curvature-Weighted Gradient Diversity (CWGD)**, a novel optimizer metric that adaptively modulates Stochastic Gradient Descent learning rates by down-weighting high-curvature directions to isolate stochastic noise.
- Derived rigorous convergence guarantees for the **CWGD-Cosine** schedule, proving a $1/(1+\alpha)$ reduction in the asymptotic convergence floor, achieving a theoretical **$2\times$ loss reduction** at the optimal $\alpha = 1$.
- Evaluated CWGD across diverse optimization regimes ($\kappa \in \{5, 10, 20, 50\}$ and varying batch sizes), consistently achieving **20% lower final suboptimality** with strong statistical significance ($p < 10^{-4}$) over 20 independent runs.
- Implemented an optimized **Hutchinson diagonal estimator** to eliminate degenerate variance ratios, achieving an exact curvature proxy with **0.5% normalized computational overhead** and open-sourced the reproducible pipeline.

Indian Institute of Management, Ranchi 🌐

Ranchi, India

Research Assistant — Multimodal Learning Systems (Supervisor: Prof. Sobhan Sarkar)

Apr 2025 – Aug 2025

- Developed a scalable, robust, end-to-end multimodal learning pipeline, integrating textual & visual data, efficiently processing **54,000+ IMDB reviews** and **30,000+ movie frames** across **10+ movies** from **4+ genres**.
- Constructed & benchmarked architectures combining **BERT embeddings**, **Vision Transformer (ViT)** features, and **CLIP-based alignment** under multiple fusion strategies for improved multimodal performance evaluation.
- Evaluated the models on **five rigorous performance metrics** (Accuracy, F1, AUC, MAP, NDCG), achieving exceptionally strong peak performance of **0.875 Accuracy** and **0.982 NDCG**, best on 'Interstellar' dataset.
- Introduced the **Dempster-Shafer evidential fusion** to explicitly model cross-modal uncertainty & belief conflict, enabling adaptive, confidence-weighted multimodal predictions improving robustness over baseline fusion methods.

University of Oulu, Finland 🌐

Oulu, Finland

Research Intern @ Efficient Neural Computation (Supervisor: Prof. Shuai Lee)

May 2025 – Jun 2025

- Formulated & solved constrained quadratic programs for pruning Spiking Neural Networks (SNNs), achieving **87.5% spatial sparsity** while preserving accuracy, using **CVXPY** with biologically inspired constraints.
- Integrated criticality-based neuron selection into pruning pipeline using layer-wise importance scores, **retaining top 112 out of 784 neurons**, and visualizing accuracy versus criticality thresholds to guide structured pruning.
- Implemented **KL-divergence-based temporal sparsity regularization**, penalizing redundant spiking across timesteps and integrating it as a weighted constraint in CVXPY to jointly reduce spatial and temporal redundancy.

Publications

- Muhammad Hamza et al.** (2025). *Movie Review Helpfulness Prediction using Semantic Alignment & Evidential Fusion-based Multimodal Approach*. Accepted at **iSAI-NLP 2025**; Published at **IEEE Xplore 2026**.
- Muhammad Hamza** and Ayush Goel. (2026). *Curvature-Weighted Gradient Diversity: A Noise Measure for Geometry-Adaptive SGD Schedules*. **arXiv:2606.30455 (Preprint)**.
- Muhammad Hamza**. (2026). *Criticality-Constrained Iterative Pruning for Energy-Efficient Spiking Neural Networks via Combined Importance Scoring*. **arXiv:2606.30676 (Preprint)**.

Selected Projects

ApexSearch: Multimodal RAG Engine for F1 Broadcasts

PyTorch, FAISS, CLIP, CLAP, LLaVA, Groq API

- Indexed **2000+** **1-FPS** video frames and **1-second audio segments** across 4 distinct F1 broadcast events, building a **cross-modal FAISS index** supporting joint audio-visual similarity queries at runtime.
- Engineered a multimodal RAG pipeline for Formula-I broadcasts, embedding **1FPS video frames & 1-second audio segments** using **CLIP & CLAP into FAISS**, enabling **sub-50ms retrieval** across acoustic race events.
- Integrated a hosted **LLaVA-based** vision-language model through the **Groq API** to generate Formula-I commentary from multimodal evidence, eliminating GPU requirements while enabling Streamlit deployment.

VictorianGPT: Domain-Specific RAG Architecture

PyTorch, QLoRA, ChromaDB, HuggingFace

- Constructed **100K+** **instruction-tuning dataset** via regex extraction and quality filtering across **50 classical novels**
- Fine-tuned **3B-parameter LLM** with QLoRA 4-bit quantization, reducing memory by 75% while maintaining stability
- Deployed **ChromaDB-backed RAG system** on HuggingFace with retrieval weighting for context-relevant generation

VLARS: Visuo-Lingual Affective Recommender

PyTorch, FAISS, DeepFace, DistilRoBERTa

- Built a multimodal user-state encoder fusing **DeepFace** micro-expression vectors and **DistilRoBERTa** sentiment embeddings into a **unified 512-dim** affective representation for downstream retrieval.
- Designed an L1-normalized late-fusion heuristic resolving cross-modal contradictions, **improving recall@5 by 23% over text-only and 18% over face-only** baselines on a 500-clip eval set.
- Implemented high-dimensional semantic search across 5,000 movies using **MiniLM embeddings & FAISS retrieval**, achieving an average end-to-end **inference latency of 180ms** on CPU.

Achievements

- **AlgoVerse AI Research Fellow 2026:** One of the 90 selected in cohort from 90,000+ applicants, working on AI research
- **Amazon ML Summer School 2026:** One of the 3,000 participants selected from over 133,000 applicants (top 2.3%)
- **Joint Entrance Examination Main 2023:** Secured an All India Rank of 3071 among 1.2 Million+ candidates
- **Joint Entrance Examination Advanced 2023:** Secured an All India Rank of 5072 among 200K+ candidates
- **Optiver Trade-athon 2026:** Rank 23/500+ in quantitative trading challenge with real-time market-making strategies
- **NK Securities Research Hackathon 2025:** Global Rank 198/6000+ & MSE 9.8×10^{-5} in implied volatility prediction
- **Convolve 4.0 (IIT Madras):** All India Rank 102 in ML/AI screening assessment from a participant pool of 10,000+

Technical Skills

- **Languages:** Python, C++, C, SQL
- **ML & DL Frameworks:** PyTorch, TensorFlow, HuggingFace Transformers
- **Generative AI:** Transformers, LLMs, RAG, QLoRA, LoRA, PEFT, LangChain, ChromaDB, FAISS, Prompt Engineering
- **Computer Vision:** CNNs, Vision Transformers (ViT), CLIP, LLaVA, OpenCV
- **Natural Language Processing:** BERT, RoBERTa, DistilBERT, Sentence Transformers, Tokenization, Text Classification
- **Specialized ML:** Multimodal Learning, Spiking Neural Networks (SNNs), Evidential Learning, Representation Learning
- **Data Science:** NumPy, Pandas, SciPy, Scikit-learn, Matplotlib
- **Deployment & MLOps:** Git, GitHub, Streamlit, Hugging Face